

DP PHYSICS SL LAB REPORT GUIDE



Mr. Mercado

Name:	Lab:	Date:
_____	_____	_____

HOW TO USE THIS GUIDE

Use this guide while writing your laboratory report by hand. For each section: 1) Read What to Include. 2) Follow the suggested structure. 3) Look at the physics example. 4) Check your work before moving to the next section.

1. RESEARCH QUESTION

The Research Question clearly states what you are investigating.

A strong Research Question identifies:

- the Independent Variable (IV)
- the Dependent Variable (DV)
- the relationship being investigated between them

Independent Variable - IV

The variable that is intentionally changed during the experiment.

ASK YOURSELF
What am I changing?

Dependent Variable - DV

The variable that is measured or observed as a result.

ASK YOURSELF
What am I measuring or observing?

Recommended Format

FORMAT
How does [Independent Variable] affect [Dependent Variable]?

Physics Example

EXAMPLE
How does ramp angle affect the acceleration of a cart moving down an incline?

Independent Variable (IV): Ramp angle (°)

Dependent Variable (DV): Acceleration of the cart (m/s²)

Check Your Research Question

- The investigation is clearly stated.
- The IV is identified.
- The DV is identified.
- The relationship between the IV and DV is clear.

2. VARIABLES

Variables are the factors in an investigation that are changed, measured, or kept the same.

A well-designed investigation should clearly identify:

- Independent Variable
- Dependent Variable
- Controlled Variables

Independent Variable - IV

The Independent Variable is the factor that you intentionally change during the experiment.

ASK YOURSELF
What am I changing?

You should also identify the different values or variations of the Independent Variable used in the investigation.

PHYSICS EXAMPLE
Independent Variable: Ramp angle (°) Variations tested: 10°, 20°, 30°

Dependent Variable - DV

The Dependent Variable is the factor that you measure or observe as a result of changing the Independent Variable.

ASK YOURSELF
What am I measuring or observing?

Always include the appropriate scientific unit when the variable is quantitative.

PHYSICS EXAMPLE
Dependent Variable: Acceleration of the cart (m/s ²) The acceleration will be measured using a motion sensor and recorded in m/s ² .

Controlled Variables

Controlled Variables are the factors that must be kept the same during every trial.

Keeping these variables constant helps make the investigation a fair test.

ASK YOURSELF
What things must stay the same so that only the Independent Variable affects the results?

Controlled Variable	How It Will Be Controlled
Ramp length	Use the same 1.0 m ramp for every trial
Cart	Use the same dynamics cart for every trial
Starting position	Release the cart from the same marked position
Release method	Release the cart from rest without pushing
Measurement method	Use the same motion sensor and analysis method for every trial

3. HYPOTHESIS

A Hypothesis is a scientifically supported prediction about what you think will happen during the investigation.

Your hypothesis should predict the relationship between the Independent Variable and the Dependent Variable.

A strong hypothesis also explains why you expect the result.

Recommended Format

FORMAT
If [Independent Variable] is changed, then [Dependent Variable] will _____ because _____.

Physics Example

EXAMPLE
If the ramp angle increases, then the acceleration of the cart will increase because the component of gravitational force parallel to the ramp becomes greater as the angle increases.

Break It Down

IF - What are you changing?

If the ramp angle increases...

THEN - What do you predict will happen to the Dependent Variable?

...then the acceleration of the cart will increase...

BECAUSE - What is the scientific reason for your prediction?

...because a steeper ramp produces a larger component of gravitational force along the ramp.

Weak Hypothesis

WEAK EXAMPLE
I think the cart will go faster on a steeper ramp.

This gives a prediction, but it does not clearly identify the relationship between the variables or explain the science.

Strong Hypothesis

STRONG EXAMPLE
If the ramp angle increases, then the acceleration of the cart will increase because the component of gravity acting parallel to the ramp increases, producing a greater net force on the cart.

4. BACKGROUND INFORMATION

The Background Information explains the science needed to understand your investigation.

Use the class textbook as your main source of scientific information.

Your Background Information should contain two paragraphs.

Paragraph 1 - Purpose of the Investigation

- What are you investigating?
- What is the purpose of the experiment?
- Why are you collecting this information?

SENTENCE STARTER

The purpose of this investigation is to determine how _____ affects _____.

PHYSICS EXAMPLE

The purpose of this investigation is to determine how ramp angle affects the acceleration of a cart moving down an incline. Different ramp angles will be tested and the acceleration of the cart will be measured.

Paragraph 2 - Scientific Information About the IV, DV, and Their Relationship

- What is my IV?
- What does it mean scientifically?
- What important information should someone know about it?

PHYSICS EXAMPLE

Ramp angle is the angle between the surface of an incline and the horizontal. As the angle increases, a greater component of the gravitational force acts parallel to the ramp and pulls the cart downhill.

Dependent Variable (DV)

- What is my DV?
- What exactly am I measuring?
- What scientific process does it represent?

PHYSICS EXAMPLE

Acceleration is the rate at which velocity changes with time. In this investigation, the acceleration of the cart as it moves down the ramp is measured in meters per second squared (m/s^2) using a motion sensor.

Relationship Between the IV and DV

- Explain scientifically how or why the IV may affect the DV.
- Connect the scientific information about the IV and DV.

SENTENCE STARTER

As _____ changes, _____ may change because _____.

PHYSICS EXAMPLE

As ramp angle increases, the cart may accelerate more quickly. A steeper incline increases the component of gravitational force parallel to the ramp, which can produce a greater acceleration of the cart.

5. MATERIALS

List everything that was actually used during the investigation.

Your materials should include:

- materials
- equipment
- quantities
- sizes, amounts, or measurements
- appropriate scientific units

Be specific.

Weak Example

WEAK EXAMPLE
Ramp Cart Sensor Protractor

This does not provide enough information.

Better Physics Example

- 1 dynamics ramp (1.0 m)
- 1 dynamics cart
- 1 motion sensor
- 1 protractor (°)
- 1 meter stick (m)
- 1 data-logging interface
- 1 stop block

6. PROCEDURE

The Procedure explains exactly how the investigation was completed.

Someone who reads your procedure should be able to understand what you did and repeat the experiment.

Your procedure should be:

- written in numbered steps
- clear
- organized
- specific
- written in the correct order
- detailed enough to repeat the investigation

Your Procedure Should Explain

- how the equipment was set up
- how the Independent Variable was changed
- how the Dependent Variable was measured
- how Controlled Variables were kept constant
- how many trials were completed
- when measurements were taken
- important safety precautions when necessary

Physics Example

1. Set up a 1.0 m dynamics ramp with one end raised.
2. Measure the ramp angle with a protractor and set it to 10°.
3. Place a motion sensor at the top of the ramp behind the cart.
4. Place the cart at the same marked starting position on the ramp.
5. Release the cart from rest without pushing it.
6. Start data collection at the same moment the cart is released.
7. Allow the cart to roll freely down the ramp.
8. Use the velocity-time graph to determine the cart acceleration.
9. Record the acceleration in m/s² in the Raw Data table.
10. Repeat the procedure two more times at 10° for a total of three trials.
11. Repeat Steps 1-10 using ramp angles of 20° and 30°.
12. Record all measurements and observations in the Raw Data table.

Notice What the Procedure Includes

Part	Example
Independent Variable	Ramp angle changes: 10° -> 20° -> 30°
Dependent Variable	Cart acceleration is measured in m/s ²
Controlled Variables	Same ramp, same cart, same starting point, same release method, same measurement method
Trials	Each ramp angle is tested three times

Weak Procedure

WEAK EXAMPLE
1. Get a ramp. 2. Put a cart on it. 3. Time it. 4. Repeat.

This procedure does not provide enough information for another person to repeat the investigation accurately.

Better Procedure Writing

BETTER EXAMPLE
Instead of: Put the cart on the ramp. Write: Place the cart at the marked starting position on a ramp set to 10°. Instead of: Measure the motion. Write: Start data collection when the cart is released and determine acceleration from the velocity-time graph.

Procedure Writing Tips

Use Numbered Steps: Write 1, 2, 3, 4... Do not write the procedure as one large paragraph.

Include Measurements: Use specific values such as 25 mL, 5.0 g, 30 °C, or 60 s.

Include Scientific Units: Always include appropriate units with measurements.

Explain Repeated Trials: If the investigation has three trials, clearly state when and how the trials are repeated.

Keep the Procedure in Order: Write each step in the exact sequence in which it should be performed.

Safety

If the investigation involves moving objects, ramps, electrical equipment, heat, or lasers, include appropriate safety precautions.

SAFETY EXAMPLE
Secure the ramp, keep hands clear of the moving cart, and use a stop block at the end of the ramp.

7. RAW DATA

Raw Data are the original measurements or observations collected directly during the experiment.

Raw Data have not been averaged, calculated, classified, or interpreted yet.

Your Raw Data Table Must Have

13. A descriptive title
14. The Independent Variable in the first column
15. The IV organized in ascending or logical order
16. Clearly labeled headings
17. Appropriate units
18. Clearly identified trials
19. All measurements or observations recorded

Example of Quantitative Raw Data

Table 1. Effect of Ramp Angle on Cart Acceleration

Ramp Angle (°)	Trial 1 Accel. (m/s ²)	Trial 2 Accel. (m/s ²)	Trial 3 Accel. (m/s ²)
10	0.84	0.86	0.85
20	1.67	1.72	1.68
30	2.47	2.52	2.48

Notice

- The Independent Variable is in the first column.
- The ramp angles are organized from 10° -> 20° -> 30°.
- Each trial has its own column.
- The unit m/s² is included in the heading.

Qualitative Raw Data

Not every investigation produces only numbers. You may also record qualitative observations.

- direction of motion
- vibration
- sound produced
- visible deformation
- change in brightness
- motion pattern

Example of Qualitative Data

Sample	Observation Before	Observation After
A	Cart at rest	Cart sped up downhill
B	Spring at equilibrium	Spring began oscillating
C	Bulb off	Bulb illuminated

Record what you observe, not what you think happened.

OBSERVATION
The cart moved farther during each equal time interval.
INTERPRETATION
The cart was accelerating down the ramp.

The first statement is an observation. The second statement is an interpretation.

8. PROCESSED DATA

Processed Data are created when you transform or analyze your Raw Data.

Depending on the investigation, this may include:

- calculations
- averages
- comparisons
- classifications
- scientific interpretations of qualitative observations

Processed Data should not simply repeat the Raw Data table.

Quantitative Example

Using the Raw Data:

Ramp Angle (°)	Trial 1 (m/s ²)	Trial 2 (m/s ²)	Trial 3 (m/s ²)
10	0.84	0.86	0.85
20	1.67	1.72	1.68
30	2.47	2.52	2.48

You may calculate the average acceleration.

Sample Calculation

FOR 10°
Average = (Trial 1 + Trial 2 + Trial 3) ÷ Number of Trials Average = (0.84 + 0.86 + 0.85) m/s ² ÷ 3 Average = 0.85 m/s ²

Example Processed Data Table

Table 2. Average Cart Acceleration at Different Ramp Angles

Ramp Angle (°)	Average Acceleration (m/s ²)
10	0.85
20	1.69
30	2.49

Qualitative Processed Data

Qualitative data can also be processed. Instead of performing a mathematical calculation, you may:

- classify observations
- compare observations
- identify patterns
- scientifically interpret observations

RAW OBSERVATION
The cart moved farther during each equal time interval.
PROCESSED INTERPRETATION
The cart was accelerating, so its velocity increased with time.

9. CONCLUSION

Use the CER Format: Claim - Evidence - Reasoning.

Your conclusion must answer the Research Question using your experimental results.

C - CLAIM

What did the investigation show?

Restate the purpose or Research Question and clearly describe the main relationship, trend, or pattern.

SENTENCE STARTERS

The results show that...

As _____ increased, _____ decreased/increased.

Based on the results, _____ affected _____ by...

PHYSICS EXAMPLE

The results show that increasing ramp angle increased the acceleration of the cart.

E - EVIDENCE

What data support your claim?

Use at least two specific data values or observations from your investigation.

Do not simply say: The data support my claim. Show the actual evidence.

PHYSICS EXAMPLE

At 10° , the average acceleration was 0.85 m/s^2 , while at 30° , the average acceleration increased to 2.49 m/s^2 .

R - REASONING

Why did this happen scientifically?

Connect your evidence to scientific ideas. Use the scientific information discussed in your Background Information.

SENTENCE STARTERS

This occurred because...

Scientifically, this can be explained by...

The results can be explained by...

PHYSICS EXAMPLE

This result can be explained by forces on an incline. As ramp angle increases, the component of gravity parallel to the ramp becomes larger. With the same cart mass, the greater net force produces greater acceleration according to Newton's second law.

Complete CER Example

COMPLETE EXAMPLE

The results show that increasing ramp angle increased cart acceleration. At 10° , the average acceleration was 0.85 m/s^2 , while at 30° , the average acceleration increased to 2.49 m/s^2 . As the ramp becomes steeper, the component of gravitational force parallel to the ramp increases. With the same cart mass, a greater net force causes a greater acceleration according to Newton's second law. Therefore, the results indicate that increasing the ramp angle increases the acceleration of the cart.

10. EVALUATION AND IMPROVEMENT

In this section, you analyze a limitation or source of error in the investigation and explain how the investigation could be improved.

You must do more than simply name an error.

20. Identify the limitation or source of error.
21. Explain how it may have affected the results.
22. Propose a realistic improvement.
23. Explain how the improvement would improve the investigation.

Weak Example

WEAK EXAMPLE
Human error occurred. We should be more careful next time.

This is too general.

Strong Physics Example

Part	Example
Limitation / Source of Error	The cart may have received a small push when it was released from the starting position instead of being released from rest.
Effect on the Results	A small push would change the cart's initial velocity and could make the measured acceleration differ from the value produced by gravity along the ramp alone.
Improvement	Use a mechanical release so the cart begins each trial from rest at the same marked position without an added push.
Why This Would Improve the Investigation	A consistent release would reduce variation in the cart's initial conditions and make the acceleration measurements more comparable between trials.

Another Example

Part	Example
Limitation / Source of Error	The ramp angle may have shifted slightly during the experiment.
Effect on the Results	If the ramp support moved, the actual angle may not have remained at the intended value during every trial.
Improvement	Secure the ramp support and check the angle with a protractor before each trial to keep it as close as possible to the selected value.
Why This Would Improve the Investigation	Maintaining a constant ramp angle would provide better control of the Independent Variable and make comparisons between trials more reliable.

11. REFERENCES

Include the sources used for scientific information in your report.

Your class textbook must be included in the reference list.

References must follow APA 7th edition format.

Basic Book Reference Structure

FORMAT
Author's Last Name, Initial. (Year). Title of book. Publisher.

Example Format

EXAMPLE
Smith, J. A. (2024). Introduction to physics. Science Press.

In-Text Citation Example

Scientific information taken from a source should also be cited in your Background Information.

Parenthetical Citation: (Smith, 2024)

Narrative Citation: Smith (2024) explains that...